

MATHEUS SERRÃO UCHÔA

Full Stack Developer | Java · Spring Boot · Spring Data JPA · Spring Security · React · Microsoft Azure · Microservices · MySQL · Elasticsearch · Kafka · Terraform · CI/CD
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SUMMARY

Software Engineer with 6+ years of experience in full-stack development using Java/Spring Boot on the backend and React on the frontend, architecting scalable microservices, secure REST APIs, and deploying cloud-native applications on Microsoft Azure. Proven track record building backend services with Spring Data JPA, Spring Security, Spring Cloud, and Spring Batch; crafting modern React UIs with hooks, state management, and reusable component libraries; integrating Elasticsearch for search and analytics; leveraging Kafka/Event Hub for async event-driven messaging; and delivering infrastructure as code with Terraform alongside CI/CD pipelines via Azure DevOps and GitHub Actions. Experienced in JVM tuning, SQL optimization, MySQL schema design and indexing, and Azure security with Key Vault, RBAC, and Managed Identities. Advanced/fluent English speaker with hands-on experience in international cross-functional teams.

SKILLS

Backend / Java Spring: Java, Spring Boot, Spring Data JPA, Spring Security, Spring Cloud, Spring Batch, Spring Cache (Redis), JVM tuning, REST APIs, Microservices.

Frontend: React, Hooks (useState, useEffect, useContext, useReducer, custom hooks), State Management (Context API, Redux), Reusable Components, TypeScript, JavaScript, HTML, CSS, Material UI.

Cloud / Microsoft Azure: Azure App Services, Azure Functions, Azure Storage, Azure SQL/MySQL, Azure Key Vault, RBAC, Managed Identities, Azure DevOps.

IoT / DevOps: Terraform, Docker, Kubernetes, GitHub Actions, Azure DevOps Pipelines, CI/CD, Continuous Integration and Delivery.

Databases: MySQL (schema design, query optimization, indexing), PostgreSQL, Elasticsearch (integration and indexing), MongoDB, Redis, advanced SQL.

Messaging: Apache Kafka, Azure Event Hub, async event-driven processing.

Data / AI: Python, TensorFlow, Detectron2, AutoML, Machine Learning, OpenCV, time-series processing.

Methodologies: Agile, Scrum, Kanban, DevOps, Technical Leadership, Problem Solving.

Languages: Portuguese (native), English (advanced/fluent).

EXPERIENCE

Stone, Brazil Software Engineer - Delivered cross-platform B2B features with Kotlin Multiplatform for Android/iOS, reducing platform-specific rework by an estimated 25%. - Refined architecture and modularization strategy, improving app performance by ~30% and reducing technical debt in core modules. - Led CI/CD and Continuous Integration improvements with GitHub Actions across product, backend, and design teams, shortening release cycle time by ~20%. - Applied machine learning-driven analysis in software workflows to support data-informed product decisions.	Oct 2024 - Present
CESAR (Motorola Project), Manaus, AM Android Framework Engineer - Refactored Android Open Source Project (AOSP) components for framework, camera, and sensors in Java/C++, reducing frame drops in camera-heavy flows by ~18%. - Raised system performance by up to 45%, reduced crashes/ANRs by 25%, and cut response time of core services by up to 20% across production builds. - Eliminated Android version regressions in rendering, lifecycle, brightness, and hardware integration, reducing post-release defect recurrence by ~22%.	Jan 2022 - Oct 2024
Instituto de Pesquisas Eldorado, Brazil Android Platform Developer (Parallel Role) - Integrated Android Framework with Camera HAL for embedded environments, validating compatibility across 10+ sensor/device combinations. - Optimized camera and framework service flows by 20–30% and reduced crashes/ANRs by ~25% through low-level debugging in Java/C++.	Jan 2022 - Aug 2023
Federal University of Amazonas (UFAM) — IES/CETELI/LaCCAN, Brazil	Jan 2020 - Present

Full Stack Engineer / R&D Researcher (AI & Industrial Systems)

- Architected and developed backend services with **Java/Spring Boot**, implementing **Spring Data JPA** for persistence on **MySQL**, **Spring Security** for REST API authentication/authorization, and **Spring Cloud** for service configuration and discovery in microservices environments.
- Built scalable **microservices** with **Spring Boot** to ingest, process, and expose industrial data (IoT sensors, video, audio), performing **JVM tuning**, **SQL query optimization**, and **MySQL indexing** to handle high-volume time-series data workloads.
- Integrated **Elasticsearch** as the search and analytics layer over large industrial production datasets, reducing query latency by ~40% compared to direct SQL queries.
- Implemented async event processing for sensor streams using **Apache Kafka** and **Azure Event Hub**, ensuring real-time data ingestion with low latency and fault tolerance across industrial data pipelines.
- Developed responsive frontends with **React**, using **custom hooks** (useReducer, useContext, useMemo), **state management** with Context API and Redux, and **reusable component libraries** for industrial monitoring dashboards — working closely with designers to deliver clean, accessible UIs.
- Deployed cloud-native applications on **Microsoft Azure** using **Azure App Services** for web services, **Azure Functions** for serverless event processing, **Azure Storage** for artifact management, and **Azure SQL/MySQL** as managed database backend.
- Provisioned infrastructure as code with **Terraform** for Azure environments and built **CI/CD pipelines** with **Azure DevOps** and **GitHub Actions**, automating build, test, and deployment of ML pipelines and backend services.
- Configured **Azure Key Vault** for secure secrets and credentials management, applied **RBAC** and **Managed Identities** for fine-grained access control between Azure services, eliminating hardcoded credentials across research environments.
- Used **Spring Batch** for large-scale batch processing of video/sensor datasets, and **Spring Cache** with Redis to offload frequent read queries on monitoring system dashboards.
- Delivered multiple AI projects (ResNet, UNet, TSMixer, RAG benchmark, Nimrod, Prometheus) and co-authored 12+ publications, including IEEE conferences, on computer vision, audio processing, and time-series analysis.

SUPER/UFAM, Brazil

Jan 2021 - Dec 2021

Full Stack Developer

- Led full-stack delivery of an internal platform with a **Java/Spring Boot + Spring Data JPA/MySQL** backend and a **React** frontend using **hooks**, **state management**, and **reusable components**, shipping 20+ features in a single delivery cycle.
- Built secure service layers with **Spring Security** (JWT, OAuth2) and integrated **Elasticsearch** for full-text search, improving user experience when navigating large record sets.
- Implemented **CI/CD pipelines** with **GitHub Actions** and standardized API versioning, reducing rollback frequency by ~15% and cutting deployment cycle time by ~20%.
- Optimized **MySQL** queries, tuned database indexes, and applied **Spring Cache** to cut page load time by ~30% and bug/failure rate by ~20%.

COLTECH - Junior Engineering Consulting, Manaus, AM

Jan 2019 - Nov 2019

Full Stack Developer

- Developed and maintained backend services and **REST APIs** with Node.js, TypeScript, Express, Prisma, and Sequelize, serving multiple teams with responsive **React** frontends.
- Boosted backend throughput by ~30% through **SQL** optimization, **MySQL** index tuning, and server-side performance improvements.
- Instituted secure engineering and continuous integration practices with **Docker** and **CI/CD**, reducing production incidents by ~18% during traffic growth periods.

KEY PROJECTS

- **Nimrod (UFAM/TPV)**: ML pipeline with **Java/Spring Boot** backend, **MySQL**, **Elasticsearch**, and **Kafka** for data ingestion; **React** dashboard for visualization; deployed on **Azure** with **Terraform**; achieved 88% accuracy on industrial action recognition using Detectron2/AutoML.
- **Prometheus (TPV/CETELI)**: IoT sound pressure measurement system for industrial production lines using STM32, real-time signal processing, and **Kafka**-based data pipelines.
- **Kenobi (LLM Benchmark)**: RAG and LLM benchmarking pipeline with a **Spring Boot** API gateway, **Elasticsearch** as vector store, and a **React** dashboard for quality and latency metrics.

EDUCATION

Federal University of Amazonas (UFAM) — Master's Degree in Artificial Intelligence (Start: 2026)

Federal University of Amazonas (UFAM) — Bachelor's in Computer Engineering (2020 - 2024)

Fundação Matias Machline — Technical High School in Mechatronics (2017 - 2019)

ICBEU — Advanced/Fluent English (ESL) (2015 - 2019)

CERTIFICATIONS

- Embedded Android Training
- Google Machine Learning Crash Course

- Data Science Academy: Python Fundamentals
- Ezape - Entrepreneurial Virtues

SELECTED PUBLICATIONS

- Action and Assembly Time Measurement System of Industry Workers using Jetson Nano
- Action Recognition of Industrial Workers using Detectron2 and AutoML Algorithms
- Sound Pressure Level Measurement System in an Industrial Production Line using STM32 Platform
- Automatic Video Labeling with Assembly Actions of Workers on a Production Line Using ResNet
- 12+ publications in total, including IEEE conferences

TECHNICAL SKILLS (ATS)

Java, Spring Boot, Spring Data JPA, Spring Security, Spring Cloud, Spring Batch, Spring Cache, Microservices, REST APIs, JVM tuning, React, Hooks, State Management, Redux, Context API, Reusable Components, TypeScript, JavaScript, HTML, CSS, Material UI, Microsoft Azure, Azure App Services, Azure Functions, Azure Storage, Azure SQL, Azure Key Vault, RBAC, Managed Identities, Azure DevOps, Terraform, Docker, Kubernetes, GitHub Actions, CI/CD, Continuous Integration, Continuous Delivery, MySQL, Elasticsearch, PostgreSQL, MongoDB, Redis, SQL, Database Indexing, Query Optimization, Apache Kafka, Azure Event Hub, Async Messaging, Python, TensorFlow, Machine Learning, Detectron2, AutoML, OpenCV, Agile, Scrum, Kanban, DevOps, Leadership, Advanced English.